

THE EVIDENCE FOR VEGANISM

Scientific Evidence on Animals, Health, Agriculture, and the Environment

A collection of findings from peer-reviewed research, scientific institutions, and major international datasets examining vegan diets and the broader impacts of animal agriculture.

This library covers human health, nutrition, land use, water, climate, biodiversity, food production, economics, public health, animal cognition, sentience, welfare, fishing, and aquaculture.

Each statement is kept concise and paired with its source and URL so the evidence can be independently examined.

The purpose is simple:

Present the evidence clearly and let the data speak for itself.

1. LAND, CROPS & FOOD EFFICIENCY

Agriculture: ~50% of Earth's habitable land

Source: FAO data / Our World in Data

URL: <https://ourworldindata.org/land-use>

Meat + dairy: ~83% farmland | ~18% calories | ~37% protein

Source: Poore & Nemecek, Science (2018)

URL: <https://doi.org/10.1126/science.aag0216>

No animal products: ~76% less agricultural land

Source: Poore & Nemecek, Science (2018)

URL: <https://doi.org/10.1126/science.aag0216>

Global soy: ~77% animal feed | ~7% human food | ~16% industry

Source: FCRN / USDA PSD via Our World in Data

Source: Animal Equality summary of FCRN / USDA data

URL: <https://ourworldindata.org/drivers-of-deforestation>

URL: <https://animalequality.org.uk/blog/soy-and-deforestation/>

Global cereals: ~48% human food | ~41% animal feed | ~11% industrial

Source: FAO data / Our World in Data

URL: <https://ourworldindata.org/land-use-diets>

Corn: >95% of U.S. feed-grain production + use

Source: USDA Economic Research Service

URL: <https://www.ers.usda.gov/topics/crops/corn-and-other-feed-grains/>

Human-edible crops → animal feed: substantial calorie loss

Source: Cassidy et al., Environmental Research Letters (2013)

URL: <https://doi.org/10.1088/1748-9326/8/3/034015>

Direct crop consumption: more people fed from the same land

Source: Cassidy et al., Environmental Research Letters (2013)

URL: <https://doi.org/10.1088/1748-9326/8/3/034015>

Global deforestation: ~90% driven by agriculture

Source: UN FAO, Global Remote Sensing Survey

URL: <https://www.fao.org/forest-resources-assessment/remote-sensing/remote-sensing-survey/en>

Agricultural deforestation: ~50% cropland | ~38.5% grazing

Source: UN FAO, Global Remote Sensing Survey

URL: <https://www.fao.org/forest-resources-assessment/remote-sensing/remote-sensing-survey/en>

2. WATER & POLLUTION

 **Global freshwater use: Agriculture 72% | Industry 15% | Human 13%**

Source: UN FAO

URL: <https://www.fao.org/land-water/water/water-scarcity/en/>

 **Farm pollution: nitrogen + phosphorus → waterways**

Source: UN FAO

URL: <https://www.fao.org/land-water/water/water-pollution/en/>

 **Nutrient pollution: eutrophication | algal blooms | dead zones**

Source: U.S. Environmental Protection Agency

URL: <https://www.epa.gov/nutrientpollution>

 **Animal manure: nitrogen + phosphorus | runoff → waterways**

Source: U.S. Environmental Protection Agency

URL: <https://www.epa.gov/npdes/animal-feeding-operations-afos>

 **Livestock ammonia: contributes to harmful PM2.5**

Source: Domingo et al., PNAS (2021)

URL: <https://doi.org/10.1073/pnas.2013637118>

 **U.S. food air pollution: animal foods cause most health damage**

Source: Domingo et al., PNAS (2021)

URL: <https://doi.org/10.1073/pnas.2013637118>

3. CLIMATE & BIODIVERSITY

 **Food systems: ~1/3 of human-caused GHG emissions**

Source: Crippa et al., Nature Food (2021)

URL: <https://doi.org/10.1038/s43016-021-00225-9>

 **Food GHGs: ~57% animal foods | ~29% plant foods**

Source: Xu et al., Nature Food (2021)

URL: <https://doi.org/10.1038/s43016-021-00358-x>

 **Beef: ~25% of food-production GHGs**

Source: Xu et al., Nature Food (2021)

URL: <https://doi.org/10.1038/s43016-021-00358-x>

 **No animal products: ~49% lower GHGs | ~50% less acidification | ~49% less eutrophication**

Source: Poore & Nemecek, Science (2018)

URL: <https://doi.org/10.1126/science.aaq0216>

 **Agricultural expansion: habitat destruction + fragmentation**

Source: IPBES Global Assessment

URL: <https://www.ipbes.net/global-assessment>

 **50% animal-food replacement: ~31% lower farm GHGs | near-halt net forest loss by 2050**

Source: Kozicka et al., Nature Communications (2023)

URL: <https://doi.org/10.1038/s41467-023-40899-2>

4. HUMAN HEALTH & NUTRITION

 **Well-planned vegan diets: nutritionally adequate for adults**

Source: Academy of Nutrition and Dietetics (2025)

URL: <https://doi.org/10.1016/j.jand.2025.02.002>

 **Vegan diets: can support long-term cardiometabolic health**

Source: Academy of Nutrition and Dietetics (2025)

URL: <https://doi.org/10.1016/j.jand.2025.02.002>

Protein

Plant foods: all essential amino acids | vegan diets can meet protein needs

Source: Mariotti & Gardner, Nutrients (2019)

URL: <https://doi.org/10.3390/nu11112661>

Protein combining every meal: unnecessary

Source: Mariotti & Gardner, Nutrients (2019)

URL: <https://doi.org/10.3390/nu11112661>

Fiber

Fiber: plant foods ✓ | meat 0g | eggs 0g | milk 0g

Source: NIH MedlinePlus / USDA FoodData Central

URL: <https://medlineplus.gov/dietaryfiber.html>

URL: <https://fdc.nal.usda.gov/>

Cholesterol

Whole plant foods: essentially cholesterol-free

Source: USDA FoodData Central

URL: <https://fdc.nal.usda.gov/>

Animal foods: can contain dietary cholesterol

Source: USDA FoodData Central

URL: <https://fdc.nal.usda.gov/>

Randomized Vegan Trial

Vegan RCT, 8 weeks: LDL -14 mg/dL | weight -1.9 kg | fasting insulin -2.9 µIU/mL

Source: Landry et al., JAMA Network Open (2023)

URL: <https://doi.org/10.1001/jamanetworkopen.2023.44457>

Plant Protein & Longevity

Plant protein: associated with ~8% lower mortality | ~12% lower CVD mortality

Source: Naghshi et al., BMJ (2020)

URL: <https://doi.org/10.1136/bmj.m2412>

+3% energy from plant protein: associated with ~5% lower mortality

Source: Naghshi et al., BMJ (2020)

URL: <https://doi.org/10.1136/bmj.m2412>

Plant-Based Dietary Patterns

Plant-based diets: associated with ~16% lower all-cause mortality

Source: Wang et al., Nutrition Journal (2023)

URL: <https://doi.org/10.1186/s12937-023-00877-2>

Healthy plant-based diets: ~15% lower CVD | ~21% lower Type 2 diabetes

Source: Wang et al., Nutrition Journal (2023)

URL: <https://doi.org/10.1186/s12937-023-00877-2>

Cancer

Processed meat: Group 1 carcinogen | 50g/day → ~18% higher colorectal cancer risk

Source: WHO / International Agency for Research on Cancer

URL:

<https://www.who.int/news-room/questions-and-answers/item/cancer-carcinogenicity-of-the-consumption-of-red-meat-and-processed-meat>

Red meat: Group 2A probable carcinogen

Source: WHO / International Agency for Research on Cancer

URL:

<https://www.who.int/news-room/questions-and-answers/item/cancer-carcinogenicity-of-the-consumption-of-red-meat-and-processed-meat>

Group 1: evidence strength | does not mean equal danger to tobacco

Source: WHO / International Agency for Research on Cancer

URL:

<https://www.who.int/news-room/questions-and-answers/item/cancer-carcinogenicity-of-the-consumption-of-red-meat-and-processed-meat>

Meat Consumption & Common Conditions

 **Red + processed meat: +70 g/day associated with ~15% higher IHD | ~31% pneumonia | ~19% diverticular disease | ~10% colon polyps | ~30% diabetes**

Source: Papier et al., BMC Medicine (2021)

URL: <https://doi.org/10.1186/s12916-021-01922-9>

 **Poultry: +30 g/day associated with ~17% higher GERD | ~12% gastritis/duodenitis | ~10% diverticular disease | ~11% gallbladder disease | ~14% diabetes**

Source: Papier et al., BMC Medicine (2021)

URL: <https://doi.org/10.1186/s12916-021-01922-9>

 **Higher unprocessed red meat + poultry: associated with lower iron-deficiency anemia risk**

Source: Papier et al., BMC Medicine (2021)

URL: <https://doi.org/10.1186/s12916-021-01922-9>

Type 2 Diabetes

 **Higher red-meat intake: associated with higher Type 2 diabetes risk**

Source: Gu et al., American Journal of Clinical Nutrition (2023)

URL: <https://doi.org/10.1016/j.ajcnut.2023.08.021>

Modeled Global Health Effects

 **Global vegan scenario: modeled ~8.1M fewer deaths/year by 2050**

Source: Springmann et al., PNAS (2016)

URL: <https://doi.org/10.1073/pnas.1523119113>

5. PUBLIC HEALTH

 **Antibiotic misuse + overuse: accelerates antimicrobial resistance**

Source: World Health Organization

URL: <https://www.who.int/news-room/fact-sheets/detail/antimicrobial-resistance>

 **Livestock antibiotics: can select for resistant microbes**

Source: World Organisation for Animal Health

URL: <https://www.woah.org/en/what-we-do/global-initiatives/antimicrobial-resistance/>

 **Resistant microbes: animals ↔ humans ↔ ecosystems**

Source: World Health Organization

URL: <https://www.who.int/news-room/fact-sheets/detail/antimicrobial-resistance>

 **Emerging infectious diseases: majority in major study originated in animals**

Source: Jones et al., Nature (2008)

URL: <https://doi.org/10.1038/nature06536>

 **Livestock intensification: can affect zoonotic-disease risk**

Source: Gilbert et al., Animal (2021)

URL: <https://doi.org/10.1016/j.animal.2020.100123>

6. SENTIENCE & COGNITION

Sentience: capacity to experience feelings

Source: Lambert et al., Animals (2022)

URL: <https://doi.org/10.3390/ani12091182>

Fish sentience: 349 articles | 470 results | 142 species + subspecies

Source: Lambert et al., Animals (2022)

URL: <https://doi.org/10.3390/ani12091182>

Fish: pain | fear | anxiety | learning | memory

Source: Lambert et al., Animals (2022)

Source: Brown, Animal Cognition (2015)

URL: <https://doi.org/10.3390/ani12091182>

URL: <https://doi.org/10.1007/s10071-014-0761-0>

Cephalopods: octopuses | squid | cuttlefish → sentience evidence

Source: Birch et al., LSE Scientific Evidence Review (2021)

URL:

<https://www.lse.ac.uk/business/consulting/reports/review-of-the-evidence-of-sentiences-in-cephalopod-molluscs-and-decapod-crustaceans>

Decapods: crabs | lobsters | crayfish → sentience evidence

Source: Birch et al., LSE Scientific Evidence Review (2021)

URL:

<https://www.lse.ac.uk/business/consulting/reports/review-of-the-evidence-of-sentiences-in-cephalopod-molluscs-and-decapod-crustaceans>

Bivalves: nervous systems | nociception evidence | learning evidence

Source: Yurchenko et al., Frontiers in Zoology (2018)

Source: Sneddon, Journal of Experimental Biology (2015)

Source: Wageningen University & Research (2022)

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5896133/>

URL: <https://journals.biologists.com/jeb/article/218/7/967/14518/Pain-in-aquatic-animals>

URL: <https://research.wur.nl/en/publications/welfare-of-bivalves-in-aquaculture-a-first-literature-scan-and-pl/>

Clams + scallops: predator detection + escape behavior

Source: Crook & Walters, ILAR Journal (2011)

URL: <https://doi.org/10.1093/ilar.52.2.185>

Pigs: learning | memory | complex social behavior

Source: Marino & Colvin (2015)

URL: <https://escholarship.org/uc/item/8sx4s79c>

Chickens: learning | memory | social cognition

Source: Marino, Animal Cognition (2017)

URL: <https://doi.org/10.1007/s10071-016-1064-4>

Cattle: learning | memory | emotions | social relationships

Source: Marino & Allen (2017)

URL: <https://doi.org/10.26451/abc.04.01.06.2017>

7. ANIMAL WELFARE & FARMING

Farm animals: pain | fear | distress

Source: European Food Safety Authority

URL: <https://www.efsa.europa.eu/en/topics/topic/animal-welfare>

Animal transport: heat stress | thirst | fatigue

Source: European Food Safety Authority

URL: <https://www.efsa.europa.eu/en/topics/topic/animal-welfare>

Some farming systems: restrict highly motivated natural behaviors

Source: European Food Safety Authority

URL: <https://www.efsa.europa.eu/en/topics/topic/animal-welfare>

Commercial farming: can involve painful procedures

Source: European Food Safety Authority

URL: <https://www.efsa.europa.eu/en/topics/topic/animal-welfare>

8. FISHING & AQUACULTURE

Bycatch: unintended non-target animals | injury + death

Source: NOAA Fisheries

URL: <https://www.fisheries.noaa.gov/national/bycatch/understanding-bycatch>

Bycatch: turtles | marine mammals | seabirds | non-target fish

Source: NOAA Fisheries

URL: <https://www.fisheries.noaa.gov/national/bycatch/understanding-bycatch>

Aquaculture feed: wild fish | fishmeal | fish oil

Source: Global aquaculture-feed review

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11482329/>

Feed fisheries: can produce bycatch

Source: Global aquaculture-feed review

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11482329/>

9. ECONOMICS & RESOURCE USE

Plant-based diets: can cost less in high-income countries

Source: Springmann et al., Lancet Planetary Health (2021)

URL: [https://doi.org/10.1016/S2542-5196\(21\)00251-5](https://doi.org/10.1016/S2542-5196(21)00251-5)

Vegan diets: among cheapest modeled healthy diets

Source: Springmann et al., Lancet Planetary Health (2021)

URL: [https://doi.org/10.1016/S2542-5196\(21\)00251-5](https://doi.org/10.1016/S2542-5196(21)00251-5)

Global vegan scenario: modeled ~\$1.07T/year health savings by 2050

Source: Springmann et al., PNAS (2016)

URL: <https://doi.org/10.1073/pnas.1523119113>

Food-related climate damage: measurable economic costs

Source: Errickson et al., Nature Food (2021)

URL: <https://doi.org/10.1038/s43016-021-00265-1>

10. EVIDENCE KEY

~ = approximately

Associated = observational relationship, not automatic proof of causation

Modeled / scenario = estimated outcome under specified assumptions

RCT = randomized controlled trial

GHG = greenhouse gas

PM2.5 = fine particulate air pollution

CVD = cardiovascular disease

Group 1 carcinogen = sufficient evidence that an agent can cause cancer; it does not mean all Group 1 agents carry equal risk

EVIDENCE STANDARD

No claim enters this collection simply because it supports veganism.

Every factual claim has a source directly underneath it.
Every URL must lead to evidence supporting the claim.
Each fact appears once in the most appropriate category.
Related statistics are combined when they describe the same subject and can remain short without losing scientific meaning.
Complete distributions are kept on one line whenever the evidence supports the categories.
Additional supporting sources belong underneath the existing claim rather than creating a duplicate claim.
Numbers remain attached to their original population, comparison, timeframe, or model.
Associations remain associations.
Models remain models.
Claims stay short without changing their meaning.
Primary research, systematic reviews, major datasets, and authoritative scientific institutions are preferred.
Stronger evidence is used when available.
Unsupported claims are excluded.
Present the evidence clearly and let the data speak for itself.

APPENDIX A — DEFINITIONS & KEY CONCEPTS

CORE VEGANISM & ANIMAL-RIGHTS TERMS

Veganism: Seeking to avoid animal exploitation and cruelty.

Vegan: A person who practices veganism.

Plant-based: Based mainly or entirely on plant foods.

Animal: A living organism in the kingdom Animalia.

Human: An animal of the species *Homo sapiens*.

Nonhuman animal: Any animal other than a human.

Animal rights: The view that animals deserve rights and protections.

Animal welfare: An animal's physical and mental well-being.

Exploitation: Using another for one's own benefit.

Cruelty: Causing unnecessary suffering.

Abuse: Harmful or improper treatment.

Commodity: Something treated as an item for trade.

Property: Something legally owned.

Domestication: Human-directed breeding across generations.

Captivity: Being confined or controlled.

Slaughter: Killing animals, especially for food.

Factory farming: Intensive, large-scale animal farming.

Animal agriculture: Raising animals for food or other products.

Vivisection: Surgery on living animals for experimentation.

Speciesism: Unequal treatment or consideration based on species.

Carnism: The ideology supporting eating certain animals.

Anthropocentrism: Viewing humans as morally central or most important.

Anthropomorphism: Attributing human traits to nonhuman beings.

Abolitionism: Seeking to end a practice rather than regulate it.

Welfarism: Seeking better treatment rather than ending animal use.

SENTIENCE, CONSCIOUSNESS & BIOLOGY

Sentience: The capacity to experience feelings and sensations.

Consciousness: The capacity for subjective awareness or experience.

Sapience: The capacity for advanced reasoning or wisdom.

Cognition: Mental processes for acquiring and using information.

Intelligence: The capacity to learn, reason, and solve problems.

Awareness: Being conscious or perceptive of something.

Perception: Detecting and interpreting sensory information.

Sensation: An experience produced through the senses.

Emotion: A subjective feeling and physiological response.

Pain: An unpleasant sensory and emotional experience.

Suffering: An unpleasant physical or psychological experience.

Distress: Significant physical or psychological discomfort.

Pleasure: A positive or enjoyable experience.

Fear: An emotional response to perceived danger.

Stress: A response to demands or threats.

Notiception: Detection of potentially harmful stimuli.

Nociceptor: A receptor that detects potentially harmful stimuli.

Brain: An organ that processes information and coordinates behavior.

Nervous system: The body's information-processing network.

Central nervous system: The brain and spinal cord.

Neuron: A nerve cell that transmits information.

Instinct: An innate behavior not requiring prior learning.

Behavior: An organism's observable actions or responses.

Subjective experience: Experience from the individual's perspective.

Self-awareness: Awareness of oneself as an individual.

ETHICS & MORALITY

Ethics: Principles for evaluating right and wrong.

Morality: Beliefs or principles about right and wrong.

Moral: Relating to right and wrong.

Immoral: Contrary to moral principles.

Amoral: Without moral consideration.

Moral consideration: Taking another's interests into account.

Moral patient: A being whose interests deserve moral consideration.

Moral agent: A being capable of morally accountable choices.

Moral status: How much moral consideration a being receives.
Moral consistency: Applying moral principles consistently.
Ethical consistency: Applying ethical principles consistently.
Principle: A fundamental rule or belief.
Value: Something regarded as important.
Virtue: A morally valued character trait.
Compassion: Concern for suffering and a desire to relieve it.
Empathy: Understanding or sharing another's feelings.
Altruism: Concern for others without requiring personal benefit.
Harm: Physical or psychological injury or damage.
Nonmaleficence: The principle of avoiding unnecessary harm.
Beneficence: The principle of promoting well-being.
Justice: Fair treatment under relevant principles.
Fairness: Impartial and equitable treatment.
Equality: Equal status, treatment, or consideration.
Equity: Fairness accounting for relevant differences.
Autonomy: The ability to make one's own choices.
Bodily autonomy: Control over one's own body.
Consent: Voluntary agreement.
Informed consent: Voluntary agreement with adequate information.
Interest: Something affecting a being's well-being.
Well-being: A state of welfare or quality of life.
Intrinsic value: Value something has in itself.
Instrumental value: Value based on usefulness.

ETHICAL THEORIES

Utilitarianism: Ethics focused on maximizing overall well-being.
Consequentialism: Judging actions primarily by their consequences.
Deontology: Ethics based on duties, rules, and obligations.
Virtue ethics: Ethics focused on moral character.
Rights theory: Ethics focused on individual rights and protections.
Sentientism: The view that sentience warrants moral consideration.
Hedonism: The view that pleasure and suffering determine well-being.
Moral relativism: Morality depends on cultural or individual frameworks.
Moral realism: Some moral claims are objectively true.
Moral subjectivism: Moral judgments depend on individual attitudes.

RIGHTS, INTERESTS & JUSTICE

Right: A moral or legal entitlement or protection.
Legal right: A right recognized by law.
Moral right: An entitlement grounded in moral principles.
Negative right: A right against interference.
Positive right: A right requiring provision or assistance.
Discrimination: Unjustified unequal treatment based on a characteristic.
Prejudice: Judgment not adequately based on evidence.
Oppression: Systematic unjust treatment or control.
Objectification: Treating a being primarily as an object.
Dehumanization: Denying or diminishing human qualities.
Victim: Someone harmed by actions or circumstances.
Injustice: Unfair or wrongful treatment.

NECESSITY, CHOICE & RESPONSIBILITY

Necessity: Something required, not merely desired.
Need: Something required for survival, health, or a purpose.
Want: Something desired but not required.
Preference: Liking one option more than another.
Convenience: Ease requiring relatively little effort.
Survival: Continued life or existence.
Choice: Selection between available options.
Voluntary: Done by free choice.
Involuntary: Done without voluntary choice.
Responsibility: Accountability for one's actions and consequences.
Accountability: Being answerable for one's actions.
Intention: The purpose behind an action.
Intent: A person's aim in acting.
Consequence: A result of an action.

Impact: The effect of an action.

Complicity: Contributing to another's wrongdoing.

JUSTIFICATION & REASONING

Justification: A reason showing a belief or action is warranted.

Justified: Supported by sufficient reasons or evidence.

Excuse: A reason offered to reduce responsibility.

Explanation: An account of why or how something happened.

Rationalization: Making a behavior seem reasonable after the fact.

Reason: A basis for believing or doing something.

Reasoning: Drawing conclusions from information.

Logic: Principles of valid reasoning.

Argument: Reasons or evidence supporting a conclusion.

Premise: A statement supporting a conclusion.

Conclusion: A claim reached from evidence or premises.

Inference: A conclusion drawn from evidence or reasoning.

Claim: A statement asserted to be true.

Assertion: A statement presented as true.

Proposition: A statement capable of being true or false.

Validity: Whether a conclusion follows logically from its premises.

Sound argument: A valid argument with true premises.

Deduction: Reasoning where true premises guarantee the conclusion.

Induction: Reasoning from observations to a probable conclusion.

Analogy: A comparison based on relevant similarities.

Counterexample: An example disproving or limiting a general claim.

Rebuttal: Evidence or reasoning against an argument.

Refutation: Showing a claim or argument is false or unsound.

Objection: A reason against a claim or argument.

Burden of proof: The responsibility to support a claim.

CONSISTENCY, CONTRADICTIONS & HYPOCRISY

Consistency: Applying standards without contradiction.

Inconsistency: Failure to apply standards consistently.

Contradiction: Claims that cannot both be true in the same sense.

Hypocrisy: Behavior contradicting one's professed standards.

Hypocrite: Someone acting against their professed standards.

Double standard: Different standards without adequate justification.

Oxymoron: A phrase combining apparently contradictory terms.

Paradox: An apparent contradiction that may reveal a truth.

Cognitive dissonance: Discomfort from conflicting beliefs or behaviors.

Bias: A tendency that can distort judgment.

Confirmation bias: Favoring information supporting existing beliefs.

Motivated reasoning: Reasoning influenced by a desired conclusion.

COMMON LOGICAL FALLACIES

Fallacy: An error in reasoning.

Straw man: Misrepresenting an argument to attack it more easily.

Ad hominem: Attacking the person instead of the argument.

Appeal to nature: Calling something right because it is natural.

Appeal to tradition: Calling something right because it is traditional.

Appeal to popularity: Calling something true because many believe it.

Appeal to authority: Treating authority alone as proof.

Appeal to emotion: Using emotion instead of relevant evidence.

Appeal to ignorance: Treating lack of disproof as proof, or vice versa.

False dilemma: Presenting two options when more exist.

False equivalence: Treating importantly different things as equivalent.

False analogy: Using a comparison without relevant similarity.

Circular reasoning: Using a conclusion to support itself.

Begging the question: Assuming what the argument must prove.

Red herring: Distracting from the argument with an irrelevant issue.

Moving the goalposts: Changing the standard after it is met.

Special pleading: Making an unjustified exception.

Cherry-picking: Selecting favorable evidence while ignoring contrary evidence.

Hasty generalization: Generalizing from insufficient evidence.

Slippery slope: Claiming extreme consequences without adequate support.

Tu quoque: Deflecting criticism by accusing the critic.

Whataboutism: Deflecting criticism to another issue.
Non sequitur: A conclusion that does not logically follow.
Equivocation: Changing a word's meaning within an argument.
Loaded question: A question containing an unsupported assumption.
Anecdotal fallacy: Using personal experience as general proof.
Genetic fallacy: Judging a claim solely by its source.
Nirvana fallacy: Rejecting improvement because it is imperfect.
Perfect solution fallacy: Rejecting a solution because it solves too little.
Naturalistic fallacy: Deriving what ought to be from what is.

COMMON DEBATE CONCEPTS

Debate: Structured disagreement using opposing arguments.
Position: A viewpoint on an issue.
Stance: A stated position on an issue.
Definition: A statement explaining a term's meaning.
Semantics: The meaning of words and language.
Ambiguity: Wording with more than one possible meaning.
Context: Information needed to understand meaning.
Nuance: A subtle but meaningful distinction.
Evidence: Information supporting or challenging a claim.
Proof: Evidence sufficient to establish a conclusion.
Assumption: Something accepted without being established.
Presupposition: Something implicitly assumed beforehand.
Generalization: A broad claim drawn from particular cases.
Objective: Not based solely on personal feelings.
Subjective: Based on individual experience or judgment.
Fact: A claim verifiable through evidence.
Opinion: A person's belief, judgment, or view.
Belief: Something a person accepts as true.
Evidence-based: Supported by relevant evidence.

SCIENCE & EVIDENCE

Science: Systematic investigation using evidence and testing.
Scientific evidence: Evidence produced or evaluated scientifically.
Empirical evidence: Information from observation or experimentation.
Anecdote: A personal account or individual example.
Hypothesis: A testable proposed explanation or prediction.
Theory: A well-supported scientific explanation.
Scientific consensus: Broad expert agreement based on available evidence.
Peer review: Expert evaluation of scholarly work before publication.
Study: A systematic investigation of a question.
Experiment: A controlled test of a hypothesis or relationship.
Observation: Systematic recording without necessarily intervening.
Data: Recorded information collected for analysis.
Variable: A factor that can differ or change.
Control group: A comparison group without the tested intervention.
Sample: The individuals or observations studied.
Population: The broader group being studied.
Sample size: The number of observations or participants.
Relevance: The degree to which information directly bears on the question, claim, or issue being considered.
Correlation: A statistical relationship between variables.
Causation: One factor contributes to producing another.
Confounding variable: A factor that can distort an apparent relationship.
Bias in research: Systematic error that can distort results.
Replication: Repeating research to test reproducibility.
Meta-analysis: Statistical combination of multiple studies.
Systematic review: Structured review of research on a question.
Observational study: Research without assigned interventions.
Randomized controlled trial: An experiment with random group assignment.
Statistical significance: Evidence inconsistent with a specified null model.
Effect size: The magnitude of a relationship or difference.
Association: A relationship that does not itself prove causation.

FOOD, NUTRITION & ANIMAL AGRICULTURE

Omnivore: An organism able to eat plants and animals.
Herbivore: An organism adapted mainly to eating plants.

Carnivore: An organism adapted mainly to eating animals.
Diet: Foods and drinks habitually consumed.
Nutrient: A substance needed for bodily function or health.
Essential nutrient: A required nutrient the body cannot make adequately.
Protein: A nutrient made of amino acids.
Amino acid: A building block of proteins.
Essential amino acid: An amino acid that must come from diet.
Complete protein: Protein containing all essential amino acids adequately.
Fiber: Carbohydrates that resist or escape digestion.
Cholesterol: A lipid made by animals and found in animal tissues.
Saturated fat: Fat rich in saturated fatty acids.
Vitamin B12: An essential vitamin for nerves and blood cells.
Fortification: Adding nutrients to food or drink.
Supplement: A product used to add dietary nutrients or substances.
Bioavailability: How much of a substance can be absorbed and used.
Deficiency: An inadequate amount of a needed nutrient.
Malnutrition: Poor nutrition from inadequate or imbalanced intake.
Carcinogen: An agent capable of causing cancer.
Processed meat: Meat preserved or flavored by processing.

ECOLOGY & ENVIRONMENT

Environment: The conditions surrounding living organisms.
Ecosystem: Organisms and their environment interacting together.
Biodiversity: The variety of life.
Habitat: The environment where an organism lives.
Ecology: The study of organisms and their environments.
Sustainability: Meeting needs while preserving resources over time.
Deforestation: Large-scale removal of forests.
Greenhouse gas: A gas that traps heat in the atmosphere.
Methane: A potent greenhouse gas.
Carbon footprint: Greenhouse-gas emissions attributable to something.
Land use: Human management or modification of land.
Resource use: Consumption of land, water, energy, or materials.
Trophic level: An organism's position in a food chain.

USEFUL DISTINCTIONS FOR LIVE DISCUSSIONS

Sentience ≠ intelligence: Feeling and problem-solving are different capacities.
Sentience ≠ sapience: Sentience is experience; sapience is advanced reasoning.
Pain ≠ nociception: Pain is experience; nociception detects harmful stimuli.
Explanation ≠ justification: Explaining something does not make it right.
Legal ≠ ethical: Legal does not necessarily mean ethical.
Natural ≠ ethical: Natural does not necessarily mean ethical.
Tradition ≠ justification: Tradition alone does not justify a practice.
Preference ≠ necessity: Wanting something is not needing it.
Correlation ≠ causation: Association alone does not prove causation.
Anecdote ≠ scientific evidence: Personal experience alone is not scientific proof.
Animal welfare ≠ animal rights: Welfare concerns treatment; rights concern permissible use.
Plant-based ≠ vegan: Plant-based describes diet; veganism extends beyond diet.
Intelligence ≠ moral worth: Cognitive ability does not define moral worth.
Disagreement ≠ refutation: Disagreement does not prove a claim false.
Assertion ≠ evidence: Saying something is true is not evidence.
Possibility ≠ probability: Possible does not mean likely.
Hypocrisy ≠ refutation: Inconsistency does not prove an argument false.